Civic Voice — Grievance Reporting System

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***Abstract* — Public complaints often take too long to progress within government offices, partly because the same report passes through several desks and, furthermore, because updates get lost along the way. For this reason, many people feel that their concerns simply disappear. While investigating this problem, I analyzed how the latest digital tools attempt to replace these slow and disconnected steps with simpler, technology-based methods. Previous studies show that NLP (Natural Language Processing) already plays a role in interpreting written complaints, and that machine learning models help classify or evaluate them faster. Some researchers have even tested deep neural networks to review photographs or short videos that users attach as evidence. After reviewing these ideas and their technical configurations, I developed a practical model—called Civic-Voice—that brings these components together into a single approach. The goal is to show how such a system can realistically reduce complaint management times and facilitate the entire process for both citizens and the officials who must respond to them.**

Keywords*— *Full-Stack Development, Artificial Intelligence, NLP, Public Grievance Redressal, Smart Governance, Sentiment Analysis.

1. INTRODUCTION

Addressing public complaints in a clear and reliable manner is an essential part of any transparent government. People often report everyday issues—damaged streets, non-functioning streetlights, uncollected accumulating trash—and they rightly expect municipal departments to respond without unnecessary delays. However, in many areas, the process for reporting these inconveniences still relies on a mixture of dispersed applications and old paper forms. Because these systems do not communicate well with each other, residents often do not know the status of their complaint, government departments work in isolation, and the use of paper further slows down the entire workflow. Over time, this disconnection ends up weakening citizens' trust in the authorities [1], [4].

As Smart City projects expanded and Information and Communication Technologies (ICT) became more common, administrations began looking for more digital methods to process complaints [1], [5]. Early online tools did make it easier for people to submit their reports, but the verification, assignment, and follow-up stages still relied heavily on staff, who had to manually review each case. Systems like the Smart Civic Issue Reporting System [4] and the Automatic Complaint Management platform in smart cities [1] increased accessibility, although the actual speed of resolution barely improved, mainly because automation remained limited.

More recent research indicates that combining full-stack web development with Artificial Intelligence (AI) can produce much more connected complaint management platforms [2], [3], [6]. In these new approaches, Natural Language Processing (NLP) interprets the complaint descriptions, while Machine Learning (ML) models organize and prioritize them based on urgency and context [8], [10]. Deep Learning (DL) systems can also analyze photos or short videos that citizens attach as evidence [10], [11]. Together, these technologies allow complaints to be handled in a more orderly, rapid, and context-aware manner.

Projects like Civic-Fix [9], SCMSC [12], and certain works on early complaint analysis in public data systems [7] demonstrate how AI aids in decision-making and reduces previously inevitable delays. Broader research—including that from international organizations like the World Bank Group [14]—indicates that technology-supported government tends to improve clarity and foster better citizen involvement.

Even with these advances, many current systems still suffer because the user interface, data storage, and intelligent processing modules function as separate pieces rather than a cohesive platform.

To address these limitations, this paper presents a unified system called Civic-Voice, which integrates data management, automated decision support, and user interaction within a single,AI-powered framework.

This study focuses on three main objectives:

1. Examine how complaint management systems have changed with the advent of digital tools and Artificial Intelligence.
2. Propose a citizen-centric model—flexible and scalable—that allows for faster and clearer responses, without unnecessary procedures that slow down the process.

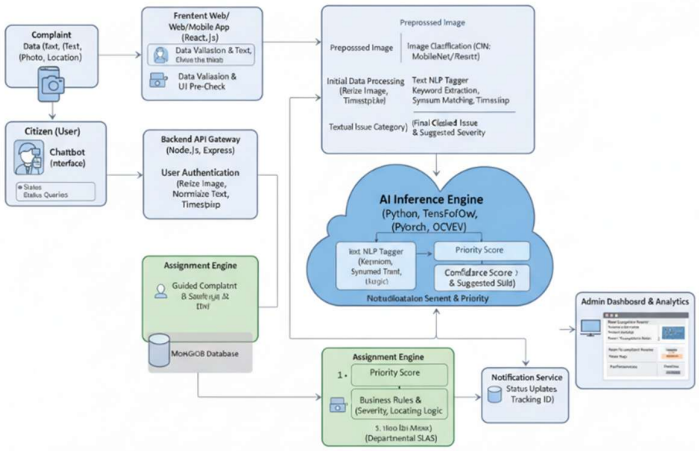

Fig 1.1 Data Flow Diagram

Figure 1.1 offers a general idea of how the public complaint tool handles user-submitted reports. Everything begins when a community member registers a problem they have observed. After that, the system takes the information from the server side and prepares it for the next steps. Next, the intelligent analysis component comes into play, which reviews the data, classifies it into different types, and also evaluates the urgency of each case. Depending on the severity of the problem and the context in which it appears, the system sends the complaint to the most appropriate team. Meanwhile, the administration dashboard shows trends, patterns, and data on team performance, and notifications keep users informed, step-by-step, through automatic updates.

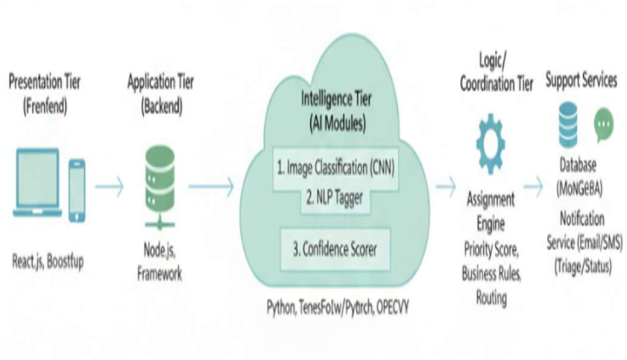

Fig 1.2: System Architecture Diagram

Figure 1.2 breaks down the functioning of Civic-Voice into several layers organized into tiers. At the top is the Presentation Layer, which is what the user directly sees, touches, and uses. Below it is the Application Layer, where verifications are executed, tasks are coordinated, and information not visible to the user is managed. Further down appears the AI Modules Layer, responsible for enhancing decisions through models and predictions. Then there is the Assignment Layer, which silently takes care of distributing reports to the areas that need to address them. All of this is held together by fundamental components: the database ensures the secure storage of information, and the alert system sends messages when necessary. Each module functions independently, so incorporating new technologies later on does not cause interruptions; improvements are simply integrated without complications.

2. LITERATURE REVIEW

The work conducted in this field has gradually changed, moving away from systems based solely on rules to approaches that employ machine learning, and more recently, deep learning models. In its early stages, most solutions relied on simple keyword matching to classify complaints and send them to the corresponding department. For example, Mane et al. [1] applied a very basic keyword method to redirect municipal issues; this approach worked well as long as the cases were simple, but it started to fail when the number of complaints increased or when the texts included more complex

Ref. No.	Study Title	Year	AI Technique / Model	Key Feature	Outcome / Accuracy
[1]	Automatic Complaint Management for Public Services in Smart City	2019	Rule-based NLP System	Keyword-based complaint routing for municipal services	Manual classification with limited scalability
[2]	AI-Based Solution to Enable Ease of Grievance Lodging and Tracking for Citizens	2025	AI-based Workflow Automation	Unified platform for multi-department complaint management	80% automation rate in issue redressal
[3]	Artificial Intelligence for Customer Complaint Management	2023	SVM and Decision Tree Classifiers	Sentiment-aware routing using text analysis	Achieved 85% classification accuracy
[4]	Smart Civic Issue Reporting System	2022	Android + IoT Integration	Mobile-based image-supported complaint registration	Medium automation, IoT data integration
[5]	Smart Maintenance System Using Augmented Reality for Urban Infrastructure Inspection	2023	AR + AI Object Detection	Augmented reality with computer vision for issue visualization	Improved maintenance response time by 30%
[6]	Automated Complaint Classification and Routing Using NLP and Machine Learning	2024	TF-IDF + Naïve Bayes	Text-based complaint classification via NLP preprocessing	90% text classification accuracy
[7]	Proactive Complaint Management in Public Sector Informatics Using AI	2025	Semantic Pattern Recognition (AI)	Context-aware complaint prediction using ontology	93% semantic matching accuracy
[8]	Automated Classification of Public Transport Complaints via Text Mining Using LLMs	2025	LLM + Embedding-based Classifier	Text-mining via large language models and embeddings	95% accuracy with contextual understanding
[9]	CivicFix: Smart Complaint Routing for Urban Solutions	2025	AI-based Routing Engine	Multi-channel data intake (IoT, SMS, web) with smart routing	92% efficient routing accuracy
[10]	Hybrid Deep Learning Models for Text Classification	2025	TriDistilBERT + BiGRU Hybrid	Deep contextual embeddings with hybrid architecture	96% accuracy with reduced false positives
[11]	Performance-Driven Hybrid Text-Image Classification Model	2025	Transformer + CNN Fusion	Multimodal fusion for text and image-based complaints	94% performance across modalities
[12]	SCMSC: A Smart Civic Complaint Management System for Cities	2025	Full-Stack AI-enabled Framework	Cloud-based ML pipeline for urban issue management	Fully automated feedback and redressal tracking
[13]	Enhancing User Experience in Full Stack Web Applications through AI & ML	2025	Predictive UX ML Models	User-interaction optimization for AI-based web portals	88% improvement in UX adaptability
[14]	Amplifying Citizens' Voice through AI-Powered Grievance Redress Systems (World Bank Group)	2025	AI Policy and Governance Models	Large-scale AI integration in public grievance systems	Improved citizen engagement and trust metrics

meanings. Over time, research such as that by Swarnkar and Rathore [2], as well as Banga and Peddireddy [3], began using supervised machine learning models, including SVM (Support Vector Machine) and decision trees. Some of these proposals also integrated sentiment analysis to evaluate how much urgency the user expressed.

3. METHODOLOGY

In Civic-Voice, the system is not made as one big program. Instead, it is divided into different parts and each part does its own work. These parts talk to each other when needed. Because of this, if something needs to be changed later, we can change only that part without touching the whole system. The main idea is to handle complaints faster and keep the user updated properly.

3.1 Overall Architecture

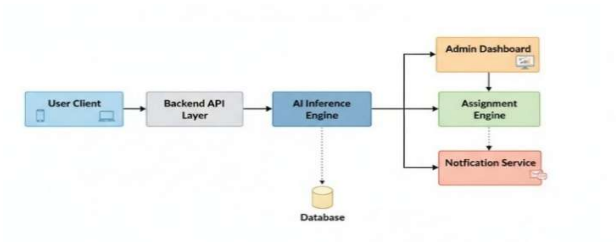

Fig 3.1: System Architecture

The system mostly includes four layers:

- The first layer show what's visible to the user using the system.
- The second layer does the actual processing.
- The third layer stores all piece of information.
- And the fourth layer contains the AI models which spot and sort out the issues.

These layers exchange information among themselves through simple APIs so that data moves easily between the sections.

3.2 Core Components

The system includes several separate parts, each of which relies on a particular technology.

Frontend: On the front side of the website system use React.js with Bootstrap. Users can sign up, log in and file complaints easily. The layout is made responsive so it works for both phone as well as laptop.

Backend: The backend uses Node.js with Express. It checks login details, applies the rules of the system, and connects with the database whenever data needs to be stored or fetched.

Database: MongoDB is used here. It keeps all complaint details, user accounts, updates from departments, and so on. Since MongoDB is document-based, it is easy to store different types of records.

AI Engine: Python is used for this part. Images uploaded by users are cleaned using OpenCV. Then models made in TensorFlow or PyTorch are used to recognize the complaint type using both text and image.

Notification Service: This small part sends SMS or email notifications so the user always knows what is happening with their complaint.

3.3 Data Transmission

The process happens like this:

- The user fills the complaint form and submits details and photo.
- The backend receives the complaint and sends the photo and text to the AI module.
- The AI checks the complaint type and how serious it is.
- The Assignment Engine decides which department should handle it.
- A tracking ID is given to the user, and the complaint also appears on the admin dashboard.
- Whenever the department updates anything, the user gets notified till the issue is solved.

3.4 AI Modules

Each of the four interconnected AI components that make up CivicVoice's intelligence has a distinct analytical function.

3.4.1 Image Classification

The uploaded images are checked using CNN based models. Sometimes MobileNet is used for faster processing and sometimes ResNet for better accuracy. Before this, OpenCV adjusts the image size and lighting so the model can understand it properly.

3.4.2 Text NLP Tagger

The text written by the user is also read by an NLP model. Important words are picked out and the meaning of the complaint is understood. Then text and image results are compared and whichever seems more clear is given more weight to finalize the category.

3.4.3 Routing Priority Computation

After the complaint type is known, the system checks where the issue is reported, how urgent it looks and whether similar complaints happened recently. Then it

assigns the complaint to the correct department.

3.4.4 Chatbot Integration

A simple chatbot is added to help users while submitting a complaint. It guides users if they find any difficulty in submit or tracking the complaint. This reduces workload on staff and helps even after office hours.
A simple chatbot is added to help users while submitting a complaint. It guides them if something is missing and can also give quick updates. This reduces workload on staff and helps even after office hours.

4. DISCUSSION AND EVALUATION

The Civic-Voice system aims to resolve several of the problems that typically arise when public complaints are managed manually. In most cases, these complaints are reviewed one by one by staff, which is not only time-consuming but can also lead to delays, misunderstandings, or even loss of information. In contrast, within Civic-Voice, many of these steps are automated from the start. Both citizens and officials can check the progress of each case at any time, making the process clearer and less time-consuming. Instead of an invisible and cumbersome flow, complaint management becomes more agile and transparent.

4.1 Advantages

The combination of artificial intelligence with web development within this system provides a series of practical improvements that were previously difficult to achieve.

- **Faster Processing:**
  Since the system analyzes complaints immediately and then sends them directly to the corresponding team, the usual delays of manual handling are eliminated. This means people do not have to wait as long to receive an initial response or an update.

- **Better Transparency:.:**
  Users keep getting SMS or email updates as their Users receive messages via SMS or email every time there is a change in the status of their complaint. This prevents them from having to guess what is happening. They can clearly see which department is resolving it, which helps strengthen trust between the public and officials.

- **Less Manual Work:**
  Staff no longer need to spend hours classifying and reviewing complaints one by one. The system automates much of that work, allowing teams to focus on resolving issues in the field instead of being constantly tied to the desk. This usually translates into quicker and more efficient responses.

- **Useful Data Insights:**
  Useful Information for Decision-Making Civic-Voice records not only the content of each complaint but also the location and type of problem. Over time, this data reveals clear patterns: areas with repeated failures or services that need greater attention. This helps departments set priorities in a more informed way and facilitates future planning.

- **Scalability and Flexibility:**
  The system is built with independent modules, so each part fulfills a specific function without fully depending on the others. If a new feature is desired in the future, it is only necessary to update the corresponding module, without altering the entire system. This allows the platform to grow and improve without restarting the project from scratch.

4.2. Evaluation Metrics

To check the performance of the system, some measurable points are used:
To check how well the system works, some measurable points are used:

- **Classification Accuracy:**
  This shows how correctly the AI recognized the complaint type. Better accuracy means more correct routing.

- **Average Response Time:**
  This is the time between submitting a complaint and receiving the initial response. A lower time means the system is performing efficiently.
  This is the time between submitting a complaint and getting the first official reply. A lower time means the process is running smoothly.

- **Complaint Resolution Rate:**
  This shows how many complaints are solved within the expected time limit. It reflects both system efficiency and department performance.

- **System Uptime and Reliability:**
  Since it is web-based, the system should be available whenever users want to submit complaints. Staying up most of the time helps keeps things working smoothly.
  Since it is web-based, the system should be available whenever users want to submit complaints. High uptime is important for reliability.

4.3. Overall Evaluation

In essence, Civic-Voice aims to make the complaint submission process much simpler for any user. Previously, most of these processes relied on paperwork and manual reviews, which were quite time-consuming and led to inevitable delays. With this system, a large part of these tasks is automated, so everything progresses more quickly. Furthermore, the user can check the status of their report at any time, without having to wait for replies or repeatedly ask for updates.

The integration of artificial intelligence allows the

system to interpret the content of the complaint by itself and send it to the corresponding department almost immediately, without the need for excessive additional verification. By neatly storing all complaints, the responsible teams can clearly identify which areas have recurring problems and, with that information, assign priorities and better plan their future actions. If the system is periodically reviewed and updated when necessary, it can remain operational for many years, helping to improve communication between citizens and authorities.

5. FUTURE SCOPE AND ENHANCEMENTS

Although the Civic-Voice system works well in its current state, there are still several opportunities to expand its capabilities in the coming years. There is still scope to make it more intuitive, faster, and more accessible for different types of users. Even small improvements can make it smarter and much more useful in the daily life of public services.

One of the most important enhancements would be to launch a mobile application compatible with Android and iOS. With features like automatic location tagging via GPS and the ability to submit reports offline, people could register a complaint from anywhere, without relying on a computer.

Another practical advancement would be to incorporate multilingual support and voice input. Many people are not comfortable writing in English, so allowing users to dictate their complaint or register it in their local language (such as Hindi, Bhojpuri, Marathi, Tamil, among others) would considerably expand access to the platform.

In the long term, the system could also be integrated with IoT sensors. For example, sensors placed in public lighting, water pipes, or trash containers could automatically detect failures—like a burnt-out lamp, a water leak, or a full container—and send a report without needing someone to do it manually. This would allow authorities to act before problems escalate.

With these improvements, Civic-Voice can evolve from being just a complaint registration system to becoming an intelligent platform for cities, capable of learning from data and anticipating solutions before problems become urgent.

6. CONCLUSIONS

The main objective of Civic-Voice is to make the complaint submission process simpler and faster for every citizen. Previously, many of these tasks relied almost entirely on manual work, which caused delays and, at times, confusion. With the incorporation of web technologies and artificial intelligence, the system can automatically classify complaints, send them to the appropriate department, and keep the user informed at every stage. This makes the entire procedure clearer and less bulky.

The combined use of text and image analysis, coupled with automatic routing, helps considerably reduce waiting times and improves process efficiency. Furthermore, the stored information allows for the detection of areas where certain problems appear frequently, which helps authorities better plan their maintenance tasks.

A central point of this project is that technology alone is not enough: it is also necessary to build trust and allow citizens to become actively involved. With future improvements—such as IoT sensors, multilingual features, and mobile applications—the system will be able to adapt even further and offer practical support in everyday situations.

In summary, Civic-Voice proposes a modern and scalable approach that can strengthen the bond between the community and local government, contributing to more agile, transparent, and well-managed services.

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